

# Survival Selection Grows a Private Kin Code, Not a Public Language

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## Abstract

Can a decodable signalling code *grow* from random weights under survival selection alone, and is what grows public or private to kin? In a minimal hand-coded 2-D lethal-vent world, tiny tied code-table agents evolve a decodable channel. Against an architecture-matched random-fitness control (differing only in the reproduction-fitness signal), selection *sharpens* rather than originates it: clonal descent alone reaches mutual intelligibility (MII) 0.1174 ( $\sim 73\times$  the  $1/625$  floor), and survival selection adds a paired win-margin of  $+0.0548$  (95% CI [0.0234, 0.0849],  $n=48$ ) at gen 28, a margin a post-hoc probe shows is a *transient acceleration*. The code stays private to kin (dominant-lineage share 99.1%,  $N_{\text{eff}}$  1.02): even under actively sustained diversity, and even when survival is made to depend on decoding non-kin, no public cross-lineage code bootstraps, a *pressure-vs-survival tension*. We also contribute a bidirectional self-audit that catches deflation as readily as inflation.

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Code & verdict artifacts: <https://github.com/yangchunxuan/grown-grounded-language> (archived at Zenodo, DOI 10.5281/zenodo.21074235); every result regenerates from committed, seeded verdict JSONs.

## The question and the world

Where does a shared signalling code come from? Most computational accounts *train* a channel in (gradient descent on a task with a differentiable channel; Foerster et al., 2016; Lazaridou et al., 2017) or *design* one in (an imposed iterated-learning transmission bottleneck; Kirby et al., 2008; Ren et al., 2020). We take up the prior question: can a decodable code *grow* from random weights under **survival selection alone**, and if it does, is it a **public** code or a code private to **kin**?

The substrate is deliberately minimal and entirely hand-coded: a  $48 \times 48$  lethal-vent world in which agents

metabolise, can die, and reproduce, and in which a single tied UnifiedIO code-table ( $4 \times 5 \times 6$ ,  $\mathcal{N}(0,1)$  init) is the only communicative apparatus. There is no gradient on communication, no oracle (agents never read world truth), and no borrowed weights. We measure mutual intelligibility (MII) behaviourally: route a referent through one agent's emit into another's decode and score exact-tuple recovery against a chance floor of  $1/625 \approx 0.0016$ . Within-founder (WF) and cross-founder (CF) MII separate a *private kin code* (WF high, CF at chance) from a *public* one (CF also above chance).

## Co-evolution is real, but decomposes and stays kin-private

The control is not communication-ablated: both arms carry the full code-table and run the identical loop, and the only manipulated variable is the reproduction-fitness signal (survival vs. uniform-random, with the reproduction RNG byte-matched). This isolates selection from *clonal descent-convergence*. The random-fitness arm also fixates to  $\sim$ one lineage, so its MII of **0.1174** ( $\sim 73\times$  floor) is the descent baseline, not zero. At gen 28 the survival arm reaches **0.17219**, a paired win-margin of  $+0.0548$ , **95% CI** [0.0234, 0.0849] ( $n=48$ ; Figure 1, panel a). Selection therefore *sharpens* a largely descent-determined channel rather than originating it.

A post-hoc probe ( $n=24$ ; gen 28/56/100/150) shows this is a *transient acceleration*: the survival-arm MII holds at  $\sim 0.16$ – $0.17$  while the control's rises ( $0.117 \rightarrow 0.138$ ) as clonal fixation accumulates decodability, so the paired margin decays from  $+0.0569$  (CI excludes zero) to a CI including zero by gen 100. The headline is therefore scoped to 28 generations.

The code is *not public*. The survival arm fixates to a single founder lineage (dominant-lineage share **99.1%**,  $N_{\text{eff}}$  **1.02**, 47/48 seeds single-lineage), so WF runs  $\approx 0.16$  ( $\sim 100\times$  chance) while CF sits at the floor. Almost every receiver an agent meets is kin, so a private code is selectively cheap (Hamilton, 1964).

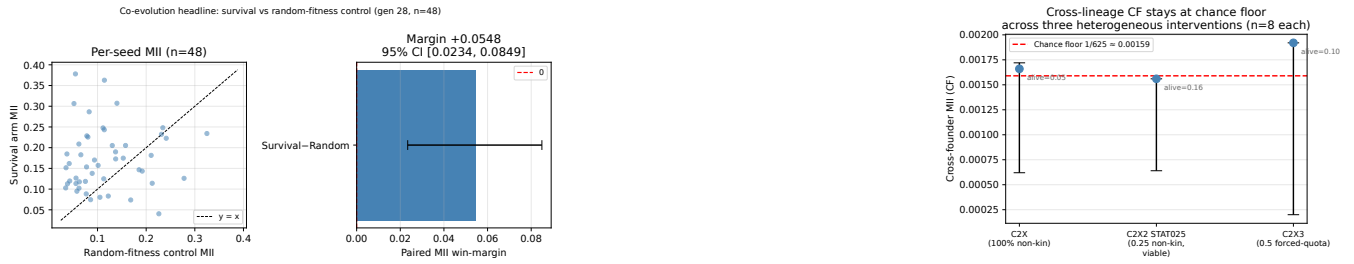


Figure 1: **A channel grows, but only kin can read it.** (a) Co-evolution headline ( $n=48$ , gen 28): the survival arm beats the architecture-matched random-fitness control by a paired MII win-margin  $+0.0548$  (95% CI  $[0.0234, 0.0849]$ ); a post-hoc probe shows this is a transient acceleration. (b) Cross-founder MII stays at the chance floor ( $\approx 0.0016$ ) across three cross-lineage interventions ( $n=8$ ); where pressure bites, survival crashes. No public code bootstraps. Sources: committed verdict JSONs.

## The wall: no public code bootstraps

Is kin-privacy merely an artifact of fast founder collapse? We sustained genuine multi-lineage coexistence by niching (tournament selection plus lineage fitness-sharing) at  $\text{pop}=96$ : the soft96 cell holds  $N_{\text{eff}} 4.10$  over 20+ coexisting generations, yet CF stays at the floor (CF 0.00153 vs. floor 0.0016; the CF–floor CI straddles zero). Maintaining diversity is *necessary but not sufficient*.

Making survival itself depend on decoding non-kin does not help either. Across three heterogeneous interventions (Figure 1, panel b: C2X2 at 0.25 non-kin stays viable (alive 0.16) yet CF 0.00156; C2X3 at 0.5 non-kin alive 0.10, CF 0.00192; C2X at 100% non-kin crashes to alive 0.046), CF never rose above chance; where the pressure bit, it crashed survival. Non-kin codes are mutually undecodable, so forced cross-listening is acting on noise, which starves foraging: a *pressure-vs-survival tension*. All cross-lineage arms are  $n=8$  (positive-or-inconclusive tier), so we report that a public code *did not* bootstrap under the regimes tested, never that none *can*.

## A bidirectional self-audit

Each claim must survive an architecture-matched control, a fair baseline,  $\geq 6$ -seed bootstrap-CI win-margins, frozen pre-registration, a no-oracle redline, and code-grounded adversarial review. Its single methodological claim is *bidirectionality*: it catches deflation as readily as inflation. Two in-scope catches show this: a pilot/formal false *negative* (a formal arm compared against a pilot control had buried a real effect, with generation count the load-bearing knob), and an RNG-offset bug in the headline control whose fix left the conclusion intact and slightly *strengthened* the margin ( $+0.045 \rightarrow +0.0548$ ).

## Contribution and scope

This separates three claims often conflated in emergent-language work: a decodable code can *evolve*, it can *matter* for survival, and it can still *fail to become public*. A decodable code grows from scratch under survival selection

but stays kin-private; selection *accelerates* a clonal-descent channel rather than originating a public one; and no public code bootstraps even under sustained diversity and direct cross-lineage pressure. The results are scoped to one hand-coded world, one architecture, and per-stage timescales (28–48 generations). We read the wall as motivating a richer world and a consequence-driven learner rather than more selection pressure. All code and seeded verdict JSONs are committed and reproducible.

## References

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